Two-step forward construction

(see Holtzman, Beauvallet, Krauth: https://arxiv.org/abs/2606.12161)

The two step forward construction uses eq. (13) to sample x_β as a Gaussian, and then given x_β together with x_0, x_τ, we construct $x_{\tau+\Delta\tau}$.

Since x_β is a Gaussian, we only need the mean and the variance. From eq. (13), there are three exponential terms that depend on x_β. We expand the terms to get the quadratic and linear terms in x_β:

In[∘]:=
```
Series[-xbeta^2/(2 sigmaR^2) - (-xbeta + xtau)^2/(2 (beta - tau)) + (-xbeta + xzero)^2/(2 beta), {xbeta, 0, 3}]
```

Out[∘]=

$$\left(-\frac{\text{xtau}^2}{2\,(\text{beta}-\text{tau})}+\frac{\text{xzero}^2}{2\,\text{beta}}\right)+\left(\frac{\text{xtau}}{\text{beta}-\text{tau}}-\frac{\text{xzero}}{\text{beta}}\right)\text{xbeta}+\left(\frac{1}{2\,\text{beta}}-\frac{1}{2\,\text{sigmaR}^2}-\frac{1}{2\,(\text{beta}-\text{tau})}\right)\text{xbeta}^2+O[\text{xbeta}]^4$$

Here we have the form $-\frac{1}{2}\,a\,x_\beta{}^2 + b\,x_\beta$, from which it follows that the standard deviation is $\frac{1}{\sqrt{a}}$ and the mean value is b/a. We can simplify the mean value a little bit:

In[∘]:=
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FullSimplify[(xtau / (beta - tau) - xzero / beta) / (1 / sigmaR^2 + 1 / (beta - tau) - 1 / beta)]
```

Out[∘]=

$$\frac{\text{sigmaR}^2\,(\text{beta}\,(\text{xtau}-\text{xzero})+\text{tau}\,\text{xzero})}{\text{beta}^2-\text{beta}\,\text{tau}+\text{sigmaR}^2\,\text{tau}}$$

After choosing x_β, we sample $x_{\tau+\Delta\tau}$ as a Gaussian bridge between x_τ and x_β. These are implemented in the program forward_construction_2.py.